

Introduction to Database Systems

Homework 1

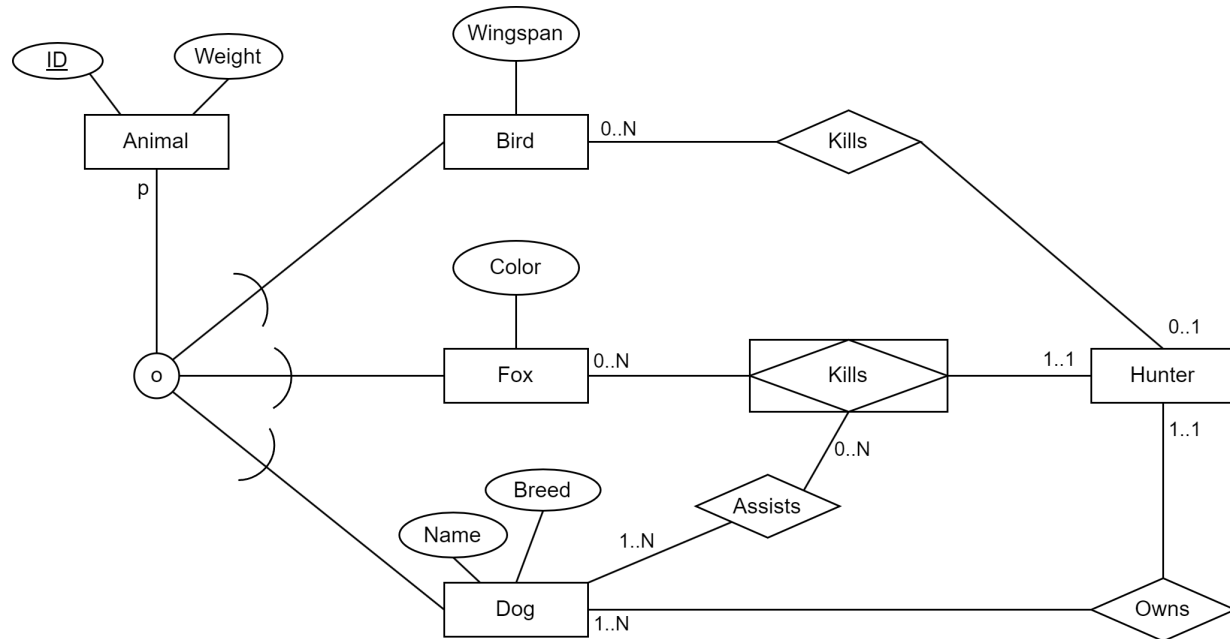
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You should submit the homework as a quiz on LearnIT. In this homework, you will practice ER model interpretation, ER model design, and translating ER models to SQL DDL.

1 ER Model Interpretation [10 Pts]

Note: This exercise is intentionally designed to be confusing, to prepare you for the real world where you may encounter counter-intuitive or outright nonsensical database designs!



1) In this part, consider ONLY the ER diagram above. Choose the correct statements.

- (a) All foxes have a weight.
- (b) A hunter can kill a fox without the assistance of a dog.
- (c) A dog can kill a bird.
- (d) Hunters are animals.
- (e) All animals are dead.
- (f) A hunter has to kill a fox with the assistance of at least one dog.
- (g) All foxes are dead.
- (h) A bird can be killed by many hunters.
- (i) All birds are dead.
- (j) Given a hunter, it is possible to find the total weight of the animals he/she has killed.
- (k) A bird can be a dog.

2 ER Model Design (org. by Hildur Davíðsdóttir) [30 Pts]

A hiking association has decided to create a database to support its operations. In collaboration with hired consultants, the association has outlined the requirements detailed below. Furthermore, they have chosen you as their database designer. Your first task is to create an ER diagram.

The following requirements were determined through interviews with the hiking association. Your database design should strictly follow the requirements definition. For example, ID columns should only be used where explicitly mentioned. When ID columns are mentioned, you can assume that they are keys.

1. The database must keep track of all people who are to be registered in the system. It should store their ID, name (divided into first and last name), address, SSN, and unique phone number.
2. In this database, each person is a hiker and/or a guide. For guides, their specialization is registered. For hikers, their highest peak is registered (if they have hiked any peaks).
3. Each guide works for at least one company. The `start_date` and `end_date` are stored for the hiring. Guides can resign from a company and later be rehired by the same company. The ID and name of companies are stored. A company can be established without having any guides working for it.
4. For guides, a list of their certifications is maintained. Each certification is identified by its name. The database also records the issue date of the certification, the name of the academy that issued it, and the corresponding grade. The grade must be assigned as a letter ranging from A to F.
5. Companies can sell products they create that have an ID and a price. All products are of either of two types: Destinations and Packages.
6. Destinations are organized journeys to a particular destination (example destination: Laugavegur). They have a name and a track name that together form a unique identifier, distance (in km), elevation (in m), and a difficulty rating. Destinations can have other prerequisite destinations, i.e. it is recommended that people have finished certain destinations before pursuing some other destination (e.g., you should have completed Preikestolen and Kjeragbolten before Trolltunga).
7. Packages are package deals that contain multiple destinations (example package: An Icelandic Hiking Adventure, containing Esjan, Vífilfell and Akrafjall). Destinations do not necessarily have to be part of any package. Packages have a name and a difficulty rating, in addition to their ID and price.

8. Hikers may join hiking clubs. Clubs have an ID, name, and date of establishment. Remember, a club is not a club if it has no members!
9. Both individual hikers and hiking clubs may be representatives of one brand. The database must keep track of these brand representatives and store their ID and the payment they receive. The database must also keep track of brands represented and should store their ID and name.
10. The database should keep track of trips. Trips have an ID and a starting date. Hikers registered in the database have all participated in at least one trip. At least four hikers must participate in each trip, and all trips must have at least one guide. Each trip has one destination, but initially, created destinations have no trips. After a trip, hikers can rate the guidance provided by each guide on the trip. The rating is stored along with a timestamp. Note that the database keeps no track of which hikers buy which products (packages/destinations), it only keeps track of the trip participation.

2) Draw an ER model reflecting the requirements above, using the ER notation of the PDBM textbook, with amendments made in the lecture. You may not extend the notation to fit the requirements; when the ER model cannot implement (part of) a requirement, it should not do so. We strongly recommend using software to create the diagram (for example, draw.io); if it is hand-drawn, please ensure it is sufficiently readable.

Using any other notation will result in 0 Pts!

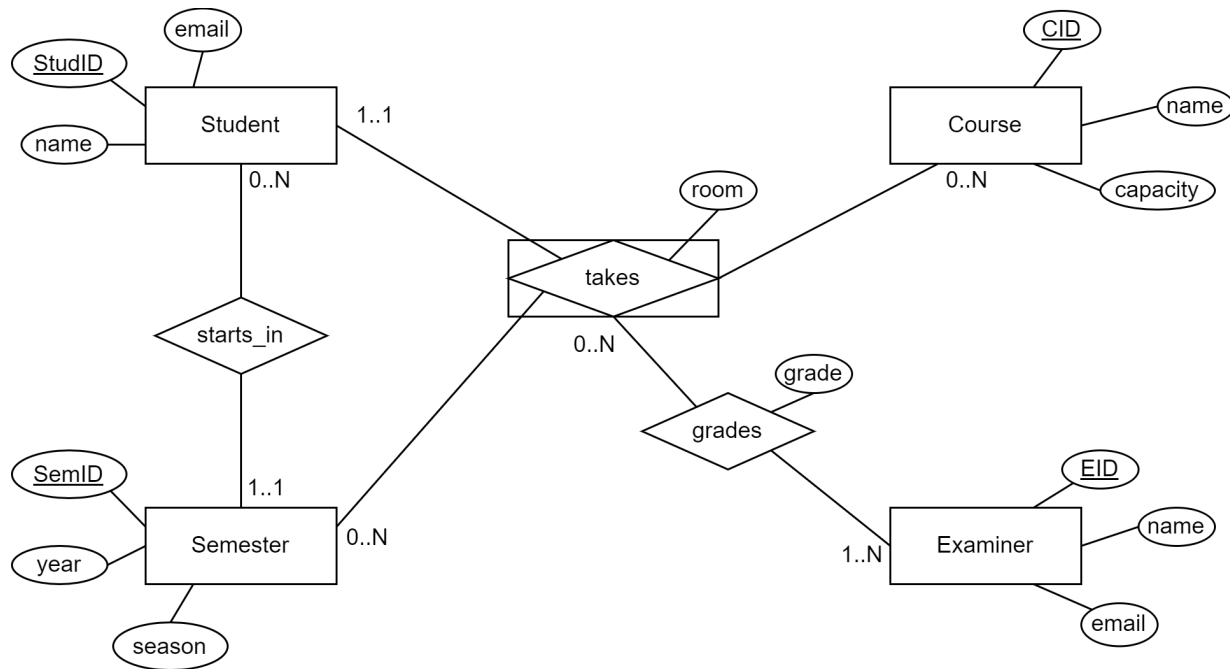


Figure 1: ER Model for a university database

3 Translating ER Model to SQL DDL [20 Pts]

Consider this ER Model for a university database along with these additional requirements:

1. StudID in Student is automatically set, starting from 1.
2. EID in Examiner is automatically set, starting from 1.
3. The season in a Semester can only have the values "Autumn" or "Spring".
4. The capacity in a Course cannot exceed 300.

3) Write SQL DDL commands to create this university database, using the methodology given in the textbook and lectures. The DDL script must run in DuckDB as a whole. The relations must include all relevant primary key, foreign key, NOT NULL and CHECK constraints.

4 Passing grade

The maximum grade is 60 points. You get 10 points for the ER Model Interpretation, 30 points for the ER Model Design, and 20 points for the Translating ER Model to SQL DDL. To pass the homework, you need to have *at least 12/60* points. However, the more you do, the more feedback you get and the more you learn.